



Let us view a graph $G = (V, E)$ as a network: its edges carry some kind of flow – of water, electricity, data or similar. How could we model this precisely?

For a start, we ought to know how much flow passes through each edge $e = xy$, and in which direction. In our model, we could assign a positive integer k to the pair (x, y) to express that a flow of k units passes through e from x to y , or assign $-k$ to (x, y) to express that k units of flow pass through e the other way, from y to x . For such an assignment $f: V^2 \rightarrow \mathbb{Z}$ we would thus have $f(x, y) = -f(y, x)$ whenever x and y are adjacent vertices.

Typically, a network will have only a few nodes where flow enters or leaves the network; at all other nodes, the total amount of flow into that node will equal the total amount of flow out of it. For our model this means that, at most nodes x , the function f will satisfy *Kirchhoff's law*

$$\sum_{y \in N(x)} f(x, y) = 0.$$

*Kirchhoff's
law*

In this chapter, we call any map $f: V^2 \rightarrow \mathbb{Z}$ with the above two properties a ‘flow’ on G . Sometimes, we shall replace $\mathbb{Z}$ with another group, and as a rule we consider multigraphs rather than graphs.¹ As it turns out, the theory of those ‘flows’ is not only useful as a model for real flows: it blends so well with other parts of graph theory that some deep and surprising connections become visible, connections particularly with connectivity and colouring problems.

¹ For consistency, we shall phrase some of our proposition for graphs only: those whose proofs rely on assertions proved (for graphs) earlier in the book. However, all those results remain true for multigraphs.

6.1 Circulations

$G = (V, E)$ Let $G = (V, E)$ be an (undirected) multigraph. Every edge $e = xy$ of G has two *directions*, (x, y) and (y, x) . (These coincide if e is a loop, so loops have only one direction.) A triple (e, x, y) consisting of an edge together with one of its directions is an *oriented edge*. The oriented edges corresponding to e are its *orientations*, denoted by $\vec{e}$ and $\bar{e}$. Thus, $\{\vec{e}, \bar{e}\} = \{(e, x, y), (e, y, x)\}$, but we cannot generally say which is which. We write

$$\vec{E} := \{ (e, x, y) \mid e \in E; x, y \in V; e = xy \}.$$

for the set of all oriented edges. We shall denote elements of $\vec{E}$ as $\vec{e}$, $\bar{e}$, etc. even if there is no previously defined edge e , and then use ‘ e ’ to refer to its underlying edge.

For an arbitrary set $\vec{F} \subseteq \vec{E}$ of oriented edges we put

$$\bar{\vec{F}} := \{ \bar{e} \mid \vec{e} \in \vec{F} \}.$$

Note that $\vec{E}$ itself is symmetrical: $\bar{\vec{E}} = \vec{E}$. For two sets $X, Y \subseteq V$ of vertices, not necessarily disjoint, and $\vec{F} \subseteq \vec{E}$, we define

$$\vec{F}(X, Y) \quad \vec{F}(X, Y) := \{ (e, x, y) \in \vec{F} \mid x \in X; y \in Y; x \neq y \},$$

$$\vec{F}(x, Y) \quad \text{abbreviate } \vec{F}(\{x\}, Y) \text{ to } \vec{F}(x, Y) \text{ etc., and write}$$

$$\vec{F}(x) \quad \vec{F}(x) := \vec{F}(x, V) = \vec{F}(\{x\}, \overline{\{x\}}).$$

$\bar{X}$ Here, as below, $\bar{X}$ denotes the complement $V \setminus X$ of a vertex set $X \subseteq V$. Note that any loops at vertices $x \in X \cap Y$ are disregarded in the definitions of $\vec{F}(X, Y)$ and $\vec{F}(x)$.

0 Let H be an abelian semigroup,² written additively with zero 0.
 f Given $X, Y \subseteq V$, not necessarily disjoint, and a function $f: \vec{E} \rightarrow H$, let

$$f(X, Y) \quad f(X, Y) := \sum_{\vec{e} \in \vec{E}(X, Y)} f(\vec{e}). \quad (1)$$

$f(x, Y)$ Instead of $f(\{x\}, Y)$ we again write $f(x, Y)$, etc.

circulation From now on, we assume that H is an abelian group. We call f a *circulation* on G (with values in H) if f satisfies the following two conditions:

- (F1) $f(e, x, y) = -f(e, y, x)$ for all $(e, x, y) \in \vec{E}$ with $x \neq y$;
- (F2) $f(v, V) = 0$ for all $v \in V$.

² This chapter contains no group theory. The only semigroups we ever consider for H are the natural numbers, the integers, the reals, the cyclic groups $\mathbb{Z}_k$, and their products $\mathbb{Z}_k \times \mathbb{Z}_m$.

If f satisfies (F1), then

$$f(X, X) = 0$$

for all $X \subseteq V$. If f satisfies (F2), then

$$f(X, V) = \sum_{x \in X} f(x, V) = 0.$$

Together, these two basic observations imply that, in a circulation, the net flow across any cut is zero:

Proposition 6.1.1. *If f is a circulation, then $f(X, \overline{X}) = 0$ for every set $X \subseteq V$.* [6.3.1]
[6.5.2]

Proof. $f(X, \overline{X}) = f(X, V) - f(X, X) = 0 - 0 = 0.$ □

Since bridges form cuts by themselves, Proposition 6.1.1 implies that circulations are always zero on bridges:

Corollary 6.1.2. *If f is a circulation and $e = xy$ is a bridge in G , then $f(e, x, y) = 0$.* □

The following lemma will be useful for induction proofs later.

Lemma 6.1.3. *Given any edge $e_0 = xy$ of G , every circulation on G/e_0 extends to a circulation on G .* [6.3.1]
[6.6.1]

Proof. Let f be a circulation on G/e_0 , and $\vec{e}_0 := (e_0, x, y)$. By the definition of multigraph minors in Chapter 1.10 we have $E(G/e_0) = E \setminus \{e_0\}$. Edges in $E \setminus E(x, y)$ are loops either in both G and G/e_0 or in neither, so the f -values on their orientations from G/e_0 are well-defined in G too. Edges $e \neq e_0$ in $E(x, y)$ are loops in G/e_0 at the contracted vertex v_{e_0} ; we take their f -value from G/e_0 as $f(\vec{e})$ for $\vec{e} := (e, x, y)$ in G , and set $f(\vec{e}) := -f(\vec{e})$ to satisfy (F1). It remains to define f on $\vec{e}_0$ and $\vec{e}_0$.

Let

$$\begin{aligned} f_x &:= \sum \{f(e, u, x) \in \vec{E} \mid u \notin \{x, y\}\} \\ f_y &:= \sum \{f(e, y, v) \in \vec{E} \mid v \notin \{x, y\}\} \\ f_{xy} &:= \sum \{f(e, x, y) \in \vec{E} \mid e \neq e_0\}. \end{aligned}$$

By (F2) for f at v_{e_0} in G/e_0 we have $f_x = f_y$. Setting $f(\vec{e}_0) := f_x - f_{xy}$ and $f(\vec{e}_0) := -f(\vec{e}_0)$ is the unique way now to extend f from $\vec{E} \setminus \{\vec{e}_0, \vec{e}_0\}$ further to a circulation on G . □

We remark that Lemma 6.1.3 has no analogue for *flows*, circulations that are non-zero everywhere (see Section 6.3): even if f is non-zero on all of G/e_0 , its extensions to G may all have to be zero on e_0 .

6.2 Flows in networks

In this section we give a brief introduction to the kind of network flow theory that is now a standard proof technique in areas such as matching and connectivity. By way of example, we shall prove a classic result of this theory, the so-called *max-flow min-cut* theorem of Ford and Fulkerson. This theorem alone implies Menger's theorem without much difficulty (Exercise 3), which indicates some of the natural power lying in this approach.

Consider the task of modelling a network with one source s and one sink t , in which the amount of flow through a given link between two nodes is subject to a certain capacity of that link. Our aim is to determine the maximum net amount of flow through the network from s to t . Somehow, this will depend both on the structure of the network and on the various capacities of its connections – how exactly, is what we wish to find out.

$G = (V, E)$
 s, t, c, N
 network
 flow

Let $G = (V, E)$ be a multigraph, $s, t \in V$ two fixed vertices, and $c: \vec{E} \rightarrow \mathbb{N}$ a map; we call c a *capacity function* on G , and the quadruple $N := (G, s, t, c)$ a *network*. Note that c is defined independently for the two orientations of an edge. A function $f: \vec{E} \rightarrow \mathbb{R}$ is a *flow* in N if it satisfies the following three conditions (Fig. 6.2.1):

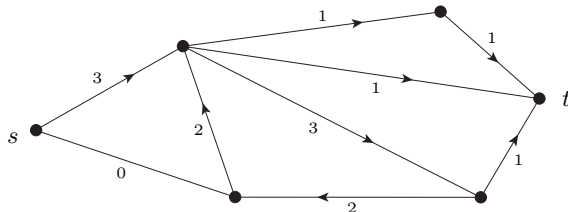


Fig. 6.2.1. A network flow in short notation: all values refer to the direction indicated (capacities are not shown)

- (F1) $f(e, x, y) = -f(e, y, x)$ for all $(e, x, y) \in \vec{E}$ with $x \neq y$;
- (F2') $f(v, V) = 0$ for all $v \in V \setminus \{s, t\}$;
- (F3) $f(\vec{e}) \leq c(\vec{e})$ for all $\vec{e} \in \vec{E}$.

integral
 f

We call f *integral* if all its values are integers.

cut in N
 capacity

Let f be a flow in N . If $S \subseteq V$ is such that $s \in S$ and $t \in \bar{S}$, we call the pair $(S, \bar{S})$ a *cut in N* , and $c(S, \bar{S})$ the *capacity* of this cut.³

Since f now has to satisfy only (F2') rather than (F2), we no longer have $f(X, \bar{X}) = 0$ for all $X \subseteq V$ (as in Proposition 6.1.1). However, the value is the same for all cuts:

³ The number $c(S, \bar{S})$ is defined in (1) of Section 6.1.